

DIVER: Diving Deeper into Distilled Data via Expressive Semantic Recovery

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Abstract

Dataset distillation aims to synthesize a compact proxy dataset that is unreadable or non-raw from the original dataset for privacy protection and highly efficient learning. However, previous approaches typically adopt a single-stage distillation paradigm, which suffers from learning specific patterns that overfit on a prior architecture, consequently suppressing the expression of semantics and leading to performance degradation across heterogeneous architectures. To address this, we propose a novel dual-stage distillation framework called **DIVER**, which leverages the pre-trained diffusion model to dive deeper into **DI**stilled data **VIA** Expressive semantic **R**ecovery, an entire process of semantic inheritance, guidance, and fusion. Semantic inheritance distills high-level semantics of abstract distilled images into the latent space to filter out architecture-specific “noise” and retain the intrinsic semantics. Furthermore, semantic guidance improves the preservation of the original semantics by directing the reverse procedure. Finally, semantic fusion is designed to provide semantic guidance only during the concrete phase of the reverse process, preventing semantic ambiguity and artifacts while maintaining the guidance information. Extensive experiments validate the effectiveness and efficiency of our method in improving classical distillation techniques and significantly improving cross-architecture generalization, requiring processing time comparable to raw DiT on ImageNet (256×256) with only 4 GB of GPU memory usage. Code is available: <https://github.com/einsteinxia/DIVER>.

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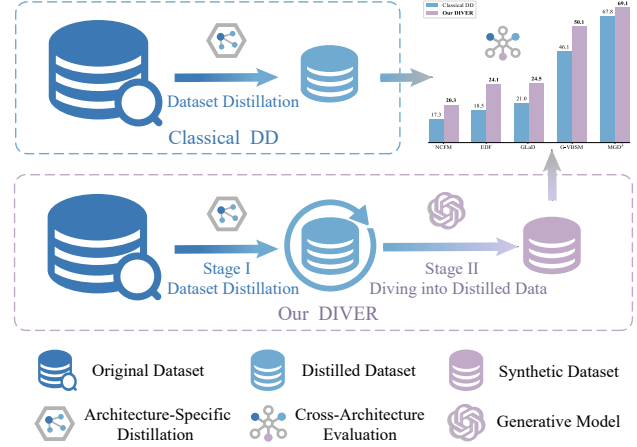


Figure 1. Comparison between classical single-stage DD and our proposed dual-stage DIVER. In stage I, DIVER is the same as classical DD. Mainly in stage II, we employ the pre-trained generative model to directly refine the distilled dataset, thereby synthesizing a new dataset termed synthetic dataset, significantly enhancing the generalization capabilities of traditional techniques across various paradigms. The quantitative results are the average performance from Tab. 1 ~ Tab. 5, respectively.

1. Introduction

Large-scale data is the fuel for deep learning, while deep learning is the engine that extracts its value, creating a virtuous cycle of mutual advancement (Fang et al., 2022; Song et al., 2025). Nevertheless, increasing data complexity introduces significant risks of private information leakage (Yu et al., 2023b) and growing computational and storage burdens on deep learning models. *Dataset Distillation* (Wang et al., 2018) (DD) mitigates these issues by compressing massive semantically rich datasets into compact, human-unreadable or non-raw representations, while preserving essential information for learning. This process not only effectively safeguards sensitive information but also substantially reduces the overhead associated with model training.

However, classical DD framework employs bi-level optimization to generate distilled datasets. In the inner loop, the network is updated to evaluate classification performance, while the outer loop synthesizes images based on certain matching strategies. For instance, gradient (Loo et al., 2023; Wang et al., 2023), distribution (Deng et al., 2024; Zhang et al., 2024), and trajectory (Du et al., 2023; Zhong et al.,

2025) matching all perform the optimization process directly in the pixel space, which tends to excessively learn specific patterns that overfit on a prior architecture (such as ConvNet) employed during distillation (Cazenavette et al., 2023). The resulting distilled images exhibit *abstract*, *noisy*, and *unrealistic* characteristics. Although such images are insightful and may enhance distillation performance, they suppress the expression of high-level visual semantics, thus falling into the cross-architecture generalization dilemma.

Some dual-time matching methods (Yin et al., 2023; Shao et al., 2024) decouple the bi-level optimization into synthesis time and training time to achieve efficient processing of large-scale datasets. As shown in Fig. 5, although this makes the distilled image appear to have some semi-clear semantic expression, the entire optimization process still operates in pixel space, and the images remain unrealistic. Recent finding (Sun et al., 2024) indicates that more realistic images facilitate cross-architecture generalization, and the performance of some existing DD methods is even worse than that of random selection (Li et al., 2025). So the generative prior methods (Cazenavette et al., 2023; Zhong et al., 2024) use GANs (Karras et al., 2019; 2020) to synthesize images optimized in latent space, its effectiveness remains constrained by expensive inner loop matching mechanisms and inadequate semi-realistic semantic representation.

Consequently, several diffusion-based approaches (Su et al., 2024; Chan-Santiago et al., 2025; Xia et al., 2026) leveraging the powerful synthesis capabilities of diffusion models have begun to emerge and demonstrated promising performance. However, they represent a decisive departure from the well-established classical DD framework, even going so far as to fully discard its conventions in favor of strictly returning to the fundamentals of coreset selection. One might whimsically ask: *Is classical DD destined for the methodological museum, or does it retain untapped potential?*

As shown in Fig. 1 (top), previous paradigms typically adopt a single-stage DD (after extracting the distilled dataset from the original dataset, it is directly used for evaluation) mode. Although obtained images may exhibit limited natural fidelity, we believe that they contain rich semantic information necessary for model generalization, it’s just that this meaningful information is suppressed by specific artifacts and unclear semantics, thus reducing generalization.

Inspired by this insight, as presented in Fig. 1 (bottom), we propose a novel dual-stage (after stage I distillation, the distilled dataset undergoes further refinement via an additional round of distillation and is subsequently evaluated) distillation framework, called DIVER. In stage I, DIVER is the same as classical DD. Mainly in stage II, we propose an innovative task *Diving into Distilled Data* (DDD), which employs the pre-trained diffusion model to progressively refine the distilled dataset via three semantic recovery strategies.

As the resolution scheme for DDD, these strategies aim to recover the expressive semantics masked and suppressed by prevalent specific patterns in the distillation output. Specifically, semantic inheritance filters out architecture-specific “noise” while distilling high-level semantic knowledge from distilled images into the latent space, ensuring that synthesized images maintain their core intrinsic semantics. Meanwhile, semantic guidance reinforces the preservation of the original semantics by steering the sampling process to produce *realistic*, *clear*, and *informative* outputs. Crucially, semantic fusion integrates conditional labels into inherited and guided latents to compensate for the lack of category information in the original features, thereby enhancing both sampling efficiency and quality.

Generally, our contributions are summarized as follows:

- We propose a dual-stage distillation framework termed DIVER, decoupling the classic dataset distillation problem into DD and DDD.
- We formally introduce DDD. To the best of our knowledge, this is the first work to straightforwardly distill knowledge from a distilled dataset into a synthetic dataset without requiring access to the original dataset.
- We propose three recovery strategies integrated into a pre-trained guided diffusion model, reviving the semantics suppressed by architecture-specific patterns in the distilled dataset without requiring additional training.
- Extensive experimental evaluations demonstrate that DIVER effectively and efficiently enhances cross-architecture generalization as a plugin in traditional DD, with only 2.48s per image on a single RTX-4090 GPU using 4GB of memory.

2. Preliminaries

2.1. Classical Dataset Distillation

For a given large-scale original dataset $\mathcal{O} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{|\mathcal{O}|}$, dataset distillation aims to build a compact distilled dataset $\mathcal{D} = \{(\tilde{\mathbf{x}}_i, \tilde{\mathbf{y}}_i)\}_{i=1}^{|\mathcal{D}|}$ that extracts rich information from $\mathcal{O}$, such that models trained on $\mathcal{D}$ achieve performance within an acceptable deviation ϵ from those trained on $\mathcal{O}$, where $|\mathcal{D}| \ll |\mathcal{O}|$ and $\mathbf{x}_i, \tilde{\mathbf{x}}_i \sim P(\mathbf{x})$, $P(\tilde{\mathbf{x}})$ are the original and distilled images with the corresponding ground-truth labels $\mathbf{y}_i, \tilde{\mathbf{y}}_i \in \mathcal{Y} = \{1, 2, \dots, C\}$, P represents data distribution, C is the number of classes. The capacity of $\mathcal{D}$ is determined by IPC (Images-Per-Class). This can be formulated as:

$$\sup_{\mathbf{x}, \tilde{\mathbf{x}} \sim P} |\mathcal{L}(\Phi_{\mathcal{O}}^p(\mathbf{x}), \mathbf{y}) - \mathcal{L}(\Phi_{\mathcal{D}}^p(\tilde{\mathbf{x}}), \tilde{\mathbf{y}})| \leq \epsilon, \quad (1)$$

where $\mathcal{L}$ denotes the loss function, $\Phi_{\mathcal{O}}^p$ and $\Phi_{\mathcal{D}}^p$ are two prior models with the same architecture but different initial parameters used during distillation.

To make this metric practically solvable, classical DD methods introduce Φ to obtain informative guidance from $\mathcal{O}$ and $\mathcal{D}$ in a chosen representation space, and iteratively optimize $\mathcal{D}$ accordingly:

$$\mathcal{D}^* = \arg \min_{\mathcal{D}} \mathcal{M}(\Phi_{\mathcal{O}}^p(x), \Phi_{\mathcal{D}}^p(\tilde{x})), \quad (2)$$

where $\mathcal{M}$ denotes various matching metrics, such as distribution matching (Loo et al., 2023), gradient matching (Wang et al., 2023), and trajectory matching (Du et al., 2023).

During distillation, Φ^p is required to be simple and efficient (e.g., ConvNet) because the optimization process is bi-level and performed directly in the pixel space, which is time-consuming and resource-intensive, and tends to excessively learn specific patterns that overfit on the prior architecture, while various complex architectures Φ^v for practical applications are preferred during evaluation.

2.2. Guided Diffusion Model

Diffusion (Ho et al., 2020; Guo et al., 2026) is a generative model that learns a mapping between Gaussian noise and the data distribution through the entire diffusion process, including a forward noising process and a reverse denoising process. For the Latent Diffusion Model (LDM) (Rombach et al., 2022), during training, the image x_0 is compressed from the pixel space $\mathcal{X}$ to the latent space $\mathcal{Z} : z_0 = \mathcal{E}(x_0)$ by using the VAE encoder $\mathcal{E}$. Then, the forward process gradually adds Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$ to z_0 : $z_t = \sqrt{\alpha_t}z_0 + \sqrt{1 - \alpha_t}\epsilon$. α_t controls the noise scale at step t .

During training, the denoising model learns to predict the noise $\epsilon_{\theta}(\hat{z}_t, t, c)$. During sampling, a Gaussian noise $\hat{z}_t$ is first initialized, the reverse process recovers the embedding $\hat{z}_0$ from ϵ_{θ} at the end, where c is a conditional input such as labels. Finally, the latent is decoded back to images: $\hat{x}_0 = \mathcal{F}(\hat{z}_0)$ by using the VAE decoder $\mathcal{F}$. For notation simplicity, the reverse process can be abstracted as: $\hat{z}_{t-1} = s(\hat{z}_t, t, \epsilon_{\theta})$, where s depends on the type of denoising. For example, in DDIM, $\hat{z}_{t-1}$ is sampled from distribution:

$$\hat{z}_{t-1} \sim \mathcal{N}(\sqrt{\alpha_{t-1}}\hat{z}_{0|t} + \sqrt{1 - \alpha_{t-1} - \sigma_t^2}\epsilon_{\theta}(\hat{z}_t, t, c), \sigma_t^2 I), \quad (3)$$

where σ_t is predefined noise factor, $\hat{z}_{0|t}$ is the clean latent predicted based on $\hat{z}_t$:

$$\hat{z}_{0|t} = \frac{1}{\sqrt{\alpha_t}}(\hat{z}_t - \sqrt{1 - \alpha_t}\epsilon_{\theta}(\hat{z}_t, t, c)). \quad (4)$$

Traditional diffusion models commonly utilize conditioning mechanisms (Ho & Salimans, 2022; Wang et al., 2022b) to generate customized outputs based on targeted user inputs, including textual prompts or categorical labels. Although these approaches work well under various constraints, they remain computationally expensive due to the need for model

retraining. Recent works (Lin et al., 2025; Bansal et al., 2023; Yu et al., 2023a) have demonstrated the effectiveness of employing frozen pre-trained diffusion models as base architectures, where the sampling process is adaptively guided by learned feedback mechanisms to produce target-specific outputs according to user requirements. The reverse process in guided diffusion can typically be implemented as follows:

$$\hat{z}_{t-1} = s(\hat{z}_t, t, \epsilon_{\theta}) - \gamma * \nabla_{\hat{z}_t} \mathcal{G}_t(\hat{z}_t), \quad (5)$$

where $\mathcal{G}$ is the guidance function used to induce the generation of customized samples, and γ is a guidance factor modulating the guidance intensity.

3. Method

The comprehensive pipeline of our proposed DIVER is illustrated in Fig. 2 and Algorithm 1 in the Appendix. In Section 3.1, we decouple the classic dataset distillation problem into DD and DDD. In Section 3.2, we use traditional DD to extract abundant information from a cumbersome original dataset into a tiny distilled dataset. In Section 3.3, we propose DDD integrating three semantic recovery strategies, which leverages the pre-trained VAE and diffusion models to filter out architecture-specific patterns in the distilled dataset and inject category information, efficiently releasing high-level expressive semantics to improve generalization.

3.1. Decoupled Dataset Distillation

An effective distilled dataset should demonstrate strong performance across various network architectures Φ^v rather than being tailored to a specific one Φ^p . In conventional DD approaches, the distilled dataset $\mathcal{D}^*$ extracted in accordance with the objective of Eqn. 2 learns specific patterns on a prior architecture Φ^p , thereby inhibiting expressive semantics that favor generalization. This outcome is attributed to the different proxy models Φ employed during distillation and evaluation, which prevented the optimization process from reaching the global optimum.

In $\mathcal{D}^*$, overfitting the specific architecture causes the distilled images to exhibit *abstract*, *noisy*, and *unrealistic* properties. They inadvertently suppress the expression of semantics, leading to a cross-architecture generalization bottleneck, which motivates us to perform in-depth exploration to mitigate the impact. As illustrated in Eqn. 6, we decouple the classic dataset distillation problem into DD (Dataset Distillation) and DDD (Diving into Distilled Data).

Define DDD. Given a distilled dataset $\mathcal{D}^*$ compressed from the original dataset $\mathcal{O}$, the goal of DDD is to refine $\mathcal{D}^*$ into a synthetic dataset $\mathcal{S}$ to mitigating the impact of the specific architecture Φ^p on generalization and minimize the objective on various architectures Φ^v :

$$\mathcal{S}^* = \arg \min_{\mathcal{S}} \mathcal{M}(\Phi_{\mathcal{O}}^v(x), \Phi_{\mathcal{S}}^v(\mathcal{H}_{\mathcal{D}^*}(\tilde{x}))) \quad s.t. \mathcal{D}^* \leftarrow \mathcal{O}, \quad (6)$$

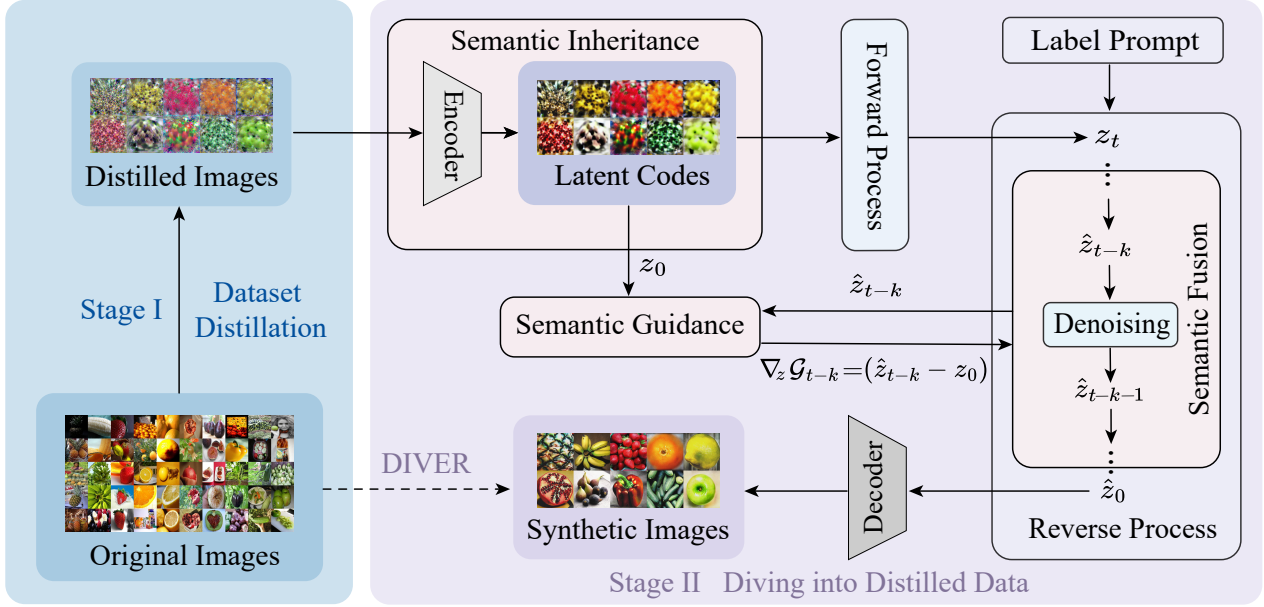


Figure 2. The overview of DIVER. Semantic inheritance filters out architecture-specific “noise” and distills high-level semantics of distilled images into the latent space, retaining the initial semantics. Semantic guidance enhances the preservation of original semantics by directing the sampling procedure to generate realistic and informative images. Semantic fusion fuses conditional labels with inherited and guided semantics only during specific stages of the reverse process, thereby enhancing both sampling efficiency and quality.

where $\mathcal{H}_{\mathcal{D}_*}(\tilde{x})$ is the synthetic image, the size $|\mathcal{S}| = |\mathcal{D}| \ll |\mathcal{O}|$. $\mathcal{H}$ denotes the semantic recovery strategies, which are designed to refine the semantics of distilled images, unlocking their suppressed potential for generalization. Generally speaking, while the initial objective of Eqn. 2 is designed for all Φ , its practical effectiveness is primarily observed at Φ^p . Our goal is to refine the distilled images to ensure their applicability at Φ^v as well.

3.2. Stage I: Dataset Distillation

Stage I comprises the entire process of classical DD, aiming to obtain the distilled dataset, applicable to most existing distillation techniques. This stage is not essential in our framework, as: (1) Our core objective is to obtain the distilled dataset rather than intermediate results from the distillation process. (2) Due to privacy reasons, the original dataset is not released by the institution. Therefore, when the distilled dataset already exists, we directly proceed to stage II without executing stage I.

3.3. Stage II: Diving into Distilled Data

3.3.1. SEMANTIC INHERITANCE

Previous studies (Cazenavette et al., 2023; Sun et al., 2024) show that images distilled by traditional methods have low-level patterns that overfit to a specific architecture. And the canonical hierarchical feature extraction principle in neural networks shows shallow layers predominantly encode low-level patterns (e.g., textures, edges, and noise)

and deeper layers capture high-level semantics (Zeiler & Fergus, 2014). We draw inspiration from LDM, and propose the semantic inheritance strategy that projects the distilled image x_0 into deep latent code z_0 via a pre-trained image encoder. Through this process, z_0 inherently inherits the high-level semantic representations from the distilled image while simultaneously filtering out architecture-specific “noise”. Then, we add noise t_f steps to z_0 :

$$z_{t_f} = \sqrt{\alpha_{t_f}} z_0 + \sqrt{1 - \alpha_{t_f}} \epsilon. \quad (7)$$

We initialize the latent code of the sampling process with $\hat{z}_{t_r} = z_{t_f}$, deliberately replacing conventional random noise initialization to preserve structural semantics throughout the diffusion process. t_r is the number of denoising steps. This operation additionally serves as a regularization mechanism, constraining the sampling trajectory to remain proximate to the initial embedding space of distilled images, thereby mitigating potential semantic drift during generation.

The choice of t_f is critical. If t_f is too large, the initial latent z_0 will be overly dominated by the Gaussian prior and thus lose the semantic information inherited from the input, making the generated results less controllable. Conversely, if t_f is too small, z_0 may significantly deviate from the assumed Gaussian distribution, violating the diffusion-model assumption and ultimately degrading the sampling stability and generation quality (Zhao et al., 2025a). An appropriate t_f should introduce moderate noise, yielding a latent that approximates a Gaussian distribution while preserving the fundamental semantic structure of the distilled data.

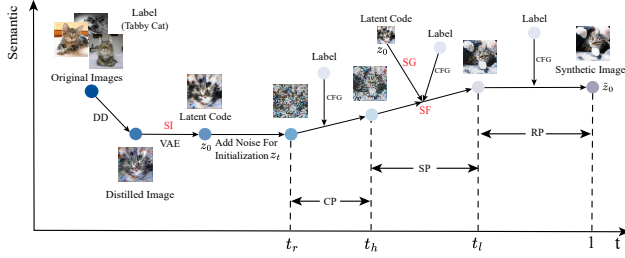


Figure 3. Semantic evolution of the entire process.

3.3.2. SEMANTIC GUIDANCE

During the image synthesis phase, our goal is to generate a high-quality image that satisfies the semantics of the specific distilled image. However, due to the continuous injection of conditional label information during the reverse process, latent code using semantic inheritance inevitably suffers from information degradation. To compensate, we introduce semantic guidance that actively reinforces semantic retention from the distilled image in the synthetic image. We design the guidance function according to Eqn. 5.

$$\mathcal{G}_t = (\hat{z}_t - z_0)^2 \cdot \sigma_t / 2, \quad (8)$$

$\mathcal{G}_t$ aims to maintain the inherited semantics of the distilled dataset so that the fused label semantics stay close to the original semantics (Chan-Santiago et al., 2025).

3.3.3. SEMANTIC FUSION

Recent studies (Yu et al., 2023a; Chen et al., 2025) have demonstrated that the sampling process of diffusion models can be divided into three distinct phases: Chaotic Phase (CP, $t_r \sim t_h$), Semantic Phase (SP, $t_h \sim t_l$), and Refinement Phase (RP, $t_l \sim 1$), with the majority of semantic content being formed during SP. Inspired by this observation, we propose to merge SG and conditional labels based on inherited latents exclusively during this critical SP. Our experiments find that this targeted approach not only enhances sampling efficiency but also improves quality.

Fig. 3 illustrates the entire semantic evolution process of our DIVER. Initially, SI inherits the valuable information from the distilled dataset. It then adds noise for t_f steps according to Eqn. 7 to initialize the latent, which preserves information while satisfying the standard diffusion mapping from a Gaussian distribution to the data distribution. Inherited semantics persist throughout the entire phase. Subsequently, Classifier-Free Guidance (CFG) is applied throughout the reverse process. Specifically, we use SG only in the SP, which involves fusing inherited semantics, conditional labels, and guidance, while using CFG only in the CP and RP. This design aims to produce clear semantics while preserving information from the distilled dataset, thus preventing semantic ambiguity and artifacts as shown in Fig. 5 (right).

4. Experiments

4.1. Experimental Setup

Datasets. Experiments are conducted on large-scale and high-resolution datasets. We evaluate the performance on the complete ImageNet-1K dataset (Deng et al., 2009) with 224×224 resolution and its twelve subsets (e.g., ImageNette, ImageA, and ImageIDC) (Howard, 2019) with 128×128 resolution. All images are resized to 256×256 in our method.

Baselines and evaluation. We compare DIVER with several state-of-the-art methods including distribution matching {DM (Zhao & Bilen, 2023), NCFM (Wang et al., 2025b)}, gradient or trajectory matching {MTT (Cazenavette et al., 2022), EDF (Wang et al., 2025a), DC (Zhao et al., 2020)}, generative prior {GLaD (Cazenavette et al., 2023)}, dual-time matching {SRe²L (Yin et al., 2023), G-VBSM (Shao et al., 2024)}, and diffusion-based {Minimax (Gu et al., 2024), D⁴M (Su et al., 2024), MGD³ (Chan-Santiago et al., 2025)} methods under the same evaluation configuration. We directly use the public distilled datasets of MTT, SRe²L, and G-VBSM, and the others use the official code to obtain the distilled datasets (NCFM w/o sampling network and using “Mix” initialization). We evaluate the methods using hard-label protocol from prior studies (Zhao & Bilen, 2023; Sajedi et al., 2023; Cazenavette et al., 2023), excluding the methods used on ImageNet-1K, which use soft-label protocol with KL divergence loss. The reported results represent averages of 3 trials on the full ImageNet-1K dataset and 5 trials on the other subsets.

Implementation details. We utilize the pre-trained Diffusion Transformer (DiT-XL/2, the image size is 256×256) and VAE model (vae-ft-mse) introduced by (Peebles & Xie, 2023), originally trained on ImageNet-1K. We use the conditioning labels and 50 sampling steps with classifier-free guidance. For the forward process, we set t_f to 25. For semantic fusion, we set t_h and t_l to 40 and 25. The scaling factor γ is set to 0.1 in all methods except NCFM, which is set to 0.02. All the experimental results of our method can be obtained on a single NVIDIA 4090 or A800 GPU.

4.2. Comparison with State-of-the-art Methods

Cross-Architecture Generalization. A critical limitation of existing DD techniques lies in their inadequate cross-architecture generalization capability, which serves as a key indicator of whether the method genuinely captures the underlying classification task semantics rather than merely overfitting to specific architectural features. As presented in Tab. 1, we show generalization results for distribution-based and trajectory-based methods with and without our DIVER. All distilled datasets generated by classic DD are distilled using a specific ConvNet. We train ResNet18, ShuffleNet-V2, MobileNet-V2, EfficientNet-B0, and ViT-b/16 (Dosovitskiy

Table 1. ImageNet Subsets (ImageFruit ~ ImageYellow, 128x128) performance on unseen architectures. The results are averaged from 5 times on the real validation sets.

Method	IPC = 1						IPC = 10					
	Fruit	Woof	Meow	Squawk	Nette	Yellow	Fruit	Woof	Meow	Squawk	Nette	Yellow
DM	11.3 $\pm$ 1.4	10.7 $\pm$ 1.1	11.7 $\pm$ 1.5	12.7 $\pm$ 1.4	13.3 $\pm$ 2.0	12.5 $\pm$ 1.5	19.3 $\pm$ 1.5	14.5 $\pm$ 0.8	17.6 $\pm$ 1.7	24.4 $\pm$ 2.3	23.7 $\pm$ 1.1	26.4 $\pm$ 1.5
Ours	18.5$\pm$1.9	13.1$\pm$0.8	14.4$\pm$1.7	18.0$\pm$1.6	21.6$\pm$2.2	22.4$\pm$1.8	22.9$\pm$1.9	16.8$\pm$1.6	20.0$\pm$1.6	28.1$\pm$3.0	28.6$\pm$2.9	29.0$\pm$2.1
NCFM	17.1 $\pm$ 1.6	12.0 $\pm$ 1.6	12.3 $\pm$ 1.5	16.8 $\pm$ 1.9	16.1 $\pm$ 1.5	17.3 $\pm$ 2.6	20.5 $\pm$ 2.5	14.3 $\pm$ 1.7	15.7 $\pm$ 2.3	22.6 $\pm$ 2.8	21.0 $\pm$ 2.6	22.5 $\pm$ 2.8
Ours	18.8$\pm$1.7	12.5$\pm$1.3	15.8$\pm$1.5	18.0$\pm$1.2	18.1$\pm$2.0	21.0$\pm$2.0	25.5$\pm$1.6	16.2$\pm$1.4	20.8$\pm$1.7	24.0$\pm$2.1	25.3$\pm$2.0	28.0$\pm$2.6
MTT	15.4 $\pm$ 1.6	13.8 $\pm$ 1.4	14.1 $\pm$ 1.7	14.0 $\pm$ 1.5	17.7 $\pm$ 1.9	17.3 $\pm$ 1.7	18.9 $\pm$ 1.4	15.9 $\pm$ 1.5	16.1 $\pm$ 1.4	21.7 $\pm$ 2.5	20.7 $\pm$ 1.8	19.1 $\pm$ 0.9
Ours	22.3$\pm$1.8	16.2$\pm$1.1	15.7$\pm$1.8	17.2$\pm$1.6	20.3$\pm$1.5	20.2$\pm$1.6	29.8$\pm$2.0	21.5$\pm$1.6	26.7$\pm$1.7	33.8$\pm$1.4	34.3$\pm$1.4	34.8$\pm$1.6
EDF	16.2 $\pm$ 1.8	15.2 $\pm$ 1.7	16.2 $\pm$ 1.6	16.5 $\pm$ 1.9	18.0 $\pm$ 1.5	18.8 $\pm$ 2.6	20.5 $\pm$ 1.5	17.2 $\pm$ 1.6	17.5 $\pm$ 1.8	23.2 $\pm$ 2.1	21.2 $\pm$ 1.8	21.1 $\pm$ 1.5
Ours	20.3$\pm$1.9	18.4$\pm$1.2	17.5$\pm$2.1	19.4$\pm$1.6	22.3$\pm$2.1	21.8$\pm$1.8	26.3$\pm$2.0	23.8$\pm$1.8	25.2$\pm$1.6	34.5$\pm$2.3	28.5$\pm$1.4	31.6$\pm$1.9

Table 2. ImageNet Subsets (ImageA ~ ImageE) performance on unseen architectures across different modes under IPC=1.

Alg.	Mode	A	B	C	D	E
DM	DD	24.0 $\pm$ 2.7	16.7 $\pm$ 2.5	18.1 $\pm$ 2.4	14.5 $\pm$ 1.7	15.4 $\pm$ 1.5
	GLaD	25.3 $\pm$ 2.1	20.1 $\pm$ 1.8	19.3 $\pm$ 1.6	18.3 $\pm$ 1.8	14.8 $\pm$ 2.6
	Ours	28.6$\pm$2.0	26.1$\pm$1.9	21.5$\pm$1.8	21.1$\pm$2.4	16.7$\pm$2.2
DC	DD	24.9 $\pm$ 3.1	21.3 $\pm$ 2.4	21.2 $\pm$ 1.5	16.9 $\pm$ 1.6	17.1 $\pm$ 2.0
	GLaD	27.4 $\pm$ 2.1	23.7 $\pm$ 1.7	22.5 $\pm$ 2.2	17.1 $\pm$ 1.1	18.6 $\pm$ 2.8
	Ours	30.5$\pm$2.2	27.8$\pm$2.1	25.1$\pm$1.8	21.4$\pm$2.0	20.6$\pm$1.6
MTT	DD	25.4 $\pm$ 2.4	21.0 $\pm$ 1.5	19.4 $\pm$ 1.7	16.1 $\pm$ 2.5	16.3 $\pm$ 2.8
	GLaD	29.1 $\pm$ 1.5	22.5 $\pm$ 3.2	19.1 $\pm$ 2.9	20.0 $\pm$ 1.4	17.1 $\pm$ 1.9
	Ours	31.6$\pm$2.7	28.7$\pm$1.8	24.9$\pm$2.3	22.1$\pm$2.2	20.2$\pm$2.1

et al., 2020) on the distilled dataset from scratch and evaluate cross-architecture generalization. The results are averaged over 5 times. Across all tested datasets, the incorporation of semantic recovery in DIVER consistently improves the performance of the generalization of all methods, with improvements ranging from marginal to substantial.

The Generative Prior. Although GLaD mitigates the architecture-specific pattern by training latent codes instead of pixel space through the traditional DD paradigm with a generator, it remains reliant on specific architectures and requires computing and storing generator gradients during training. This substantial overhead limits its scalability to high-IPC. In contrast, our method primarily builds upon the distilled dataset, requiring neither gradient computation nor access to the original dataset. Such efficiency is remarkable, and as demonstrated in Tab. 2, our approach also achieves significantly superior generalization compared to GLaD.

Dual-Time Matching. SRe²L and G-VBSM use a squeeze-recover-relabel process to decouple the bi-level optimization into synthesis time and training time, which scales to high-resolution datasets with low training cost and memory consumption. We integrate their distilled datasets into our framework to synthesize new datasets and evaluate their performance in Tab. 3. We employ ResNet18 as the recovery

Table 3. Dual-time matching methods with our DIVER on ImageNet-1K (224x224). We cite the results from G-VBSM.

Alg.	IPC	Mode	RN18	RN50	RN101
SRe ² L	10	DD	21.3 $\pm$ 0.6	28.4 $\pm$ 0.1	30.9 $\pm$ 0.1
		Ours	24.5$\pm$0.4	31.2$\pm$0.2	32.4$\pm$0.3
	50	DD	46.8 $\pm$ 0.2	55.6 $\pm$ 0.3	60.8 $\pm$ 0.5
		Ours	54.0$\pm$0.5	61.1$\pm$0.2	61.3$\pm$0.4
G-VBSM	10	DD	31.4 $\pm$ 0.5	35.4 $\pm$ 0.8	38.2 $\pm$ 0.4
		Ours	35.1$\pm$0.4	40.4$\pm$0.5	40.1$\pm$0.4
	50	DD	51.8 $\pm$ 0.4	58.7 $\pm$ 0.3	61.0 $\pm$ 0.4
		Ours	57.2$\pm$0.6	64.2$\pm$0.4	63.7$\pm$0.3

model and use soft labels to train and evaluate the ResNet- $\{18, 50, 101\}$, respectively. Our approach consistently outperforms the original method across multiple squeeze models. Notably, as the squeeze model becomes stronger (RN18 $\rightarrow$ RN101), our performance advantage gradually diminishes. This suggests a transition in the dominant factor of performance gains from the information content of the compressed data to the capability of the squeeze model trained on the original dataset. When RN101 serves as the squeeze model, the performance gap becomes marginal.

Diffusion-Based Method. D⁴M and MGD³ encode the full dataset into the latent space via a VAE for prototype learning, and expect the diffusion model to generate representative prototypes as the synthetic dataset. However, their ability to preserve prototype information remains limited. We attempt to apply their synthetic datasets, which contain relatively clear semantic structures, within our framework, but this results in performance degradation. This is mainly attributed to (1) the encoding of VAE from high-dimensional to low-dimensional space itself loses information, and (2) uncertainty introduced by diffusion. The detailed results and analysis are provided in Tab. 12 of the appendix. Therefore, we follow their paradigm by encoding the full dataset into the latent space for prototype learning, and subsequently apply our SG and SF modules to generate the synthetic dataset.

Table 4. Performance comparison over ResNet-18 with state-of-the-art DD on ImageNet-1K. Our method can be effectively integrated into both the original DiT-based and Minimax-tuned MGD³.

IPC	SRe ² L	G-VBSM	RDED	DiT	Minimax	D ⁴ M	MGD ³	MGD ³ +Ours	MGD ³ +Minimax+Ours
10	21.3 $\pm$ 0.6	31.4 $\pm$ 0.5	42.0 $\pm$ 0.1	39.6 $\pm$ 0.4	44.3 $\pm$ 0.5	27.9 $\pm$ 0.2	45.8 $\pm$ 0.3	46.4 $\pm$ 0.3	46.9$\pm$0.5
50	46.8 $\pm$ 0.2	51.8 $\pm$ 0.4	56.5 $\pm$ 0.1	52.9 $\pm$ 0.6	58.6 $\pm$ 0.3	55.2 $\pm$ 0.1	60.2 $\pm$ 0.1	60.6 $\pm$ 0.4	61.0$\pm$0.2

Table 5. Our recovery strategies are applied to the diffusion-based prototype learning. We cite the results from MGD³.

Dataset	ImageNette			ImageIDC		
IPC	10	20	50	10	20	50
D ⁴ M	59.1	64.3	70.2	52.3	55.5	62.7
D ⁴ M+Ours	62.5	65.8	75.1	53.7	58.9	67.4
MGD ³	66.4	71.2	79.5	55.9	61.9	72.1
MGD ³ +Ours	67.2	72.4	81.2	56.6	63.0	73.9

As shown in Tab. 4 and Tab. 5, our method can stably preserve the prototype information learned from the original datasets on ImageNet-1K and its subsets. Furthermore, after fine-tuning with Minimax, we achieve further performance improvement, demonstrating the superior scalability.

4.3. Ablation Studies

Effect of Each Module. As illustrated in Tab. 6, we incrementally evaluate the impact of each proposed semantic recovery component under different IPC settings. From a holistic perspective, every individual component plays a critical role in determining and enhancing the system’s overall performance. From a local perspective, the generalization of classical DD is merely comparable to random selection (14.1% vs. 15.4%), and even inferior to it under high IPC conditions (18.9% < 19.6%). Both SI and SG individually demonstrate significant performance improvements, while their combination leads to a saturation effect, resulting in only marginal additional gains. Notably, the raw DiT without any recovery modules also achieves moderate performance enhancement, which can be attributed to the fact that diffusion models enhance generalization by generating samples that closely follow the original data distribution while exhibiting greater diversity, thereby covering under-represented regions. Additionally, the inherent smoothness of their generative process yields “cleaner” samples (e.g., denoised data), which implicitly regularizes training and mitigates overfitting (Song & Ermon, 2019; Ho et al., 2020).

Additionally, While randomly selected images bring only slight improvements to our framework, they still fail to outperform DiT. This indirectly confirms that our approach relies not only on DiT, but also on the semantic information contained in the distilled images, which implicitly capture the semantics of the entire dataset. Using SF with fused semantics only in the partial phase yields considerable gains. We wil discuss the reason in Section 4.4.

Table 6. The isolated and combinatorial effects of constituent elements in our proposed semantic recovery. Random* indicates randomly selected images that are incorporated into our framework. The results are on the ImageFruit and the Alg. is MTT.

Mode	SI	SG	SF	IPC=1	IPC=10
Random	-	-	-	14.1 $\pm$ 1.4	19.6 $\pm$ 1.8
DD	-	-	-	15.4 $\pm$ 1.6	18.9 $\pm$ 1.4
Random*	✓	✓	✓	14.6 $\pm$ 1.8	21.7 $\pm$ 1.6
	×	×	×	17.8 $\pm$ 1.2	23.4 $\pm$ 1.3
	✓	×	×	19.5 $\pm$ 1.4	26.2 $\pm$ 1.5
	×	✓	×	20.4 $\pm$ 1.7	27.8 $\pm$ 1.9
	✓	✓	×	21.1 $\pm$ 1.9	28.3 $\pm$ 1.6
	✓	✓	✓	22.3$\pm$1.8	29.8$\pm$2.0

Guidance Factor. As presented in Fig. 4 (left), when the guidance factor is too small, semantic guidance becomes insufficient. Conversely, when the guidance factor is excessively high, the sampling process over-relies on the latent codes inherited from the distilled dataset, thereby diminishing the conditioning effect of the label guidance. This scenario is particularly catastrophic for the synthetic dataset, leading to a significant degradation in performance.

Forward Steps. As presented in Fig. 4 (medium), we apply different forward steps to the inherited latent. As the number of noise-adding steps increases, performance begins to improve, reaching its optimal level at 25 steps. However, further increasing the steps results in a decline in performance, as the latent variables gradually approach a pure Gaussian distribution, causing the loss of original features. Overall, an intermediate number of steps achieves a balance between initial Gaussian distribution alignment and feature preservation, leading to superior performance.

Robustness across Different Diffusion Models. As evaluated in Tab. 7, our method demonstrates consistent performance gains across different diffusion models, including Stable Diffusion V-1.5 (SD-V1.5) (Rombach et al., 2022), DiT (Peebles & Xie, 2023), and SiT (Ma et al., 2024), highlighting the robustness of DIVER. The smaller gain with SD-V1.5 is attributed to architectural discrepancies (U-Net vs. Transformer) and its different pre-training dataset (not ImageNet-1K) in contrast to the more substantial improvements observed with DiT (diffusion-based) and SiT (flow-based). In addition, this still demonstrates that our method does not necessarily rely on “data leakage”, and the priors provided by standard diffusion models remain effective.

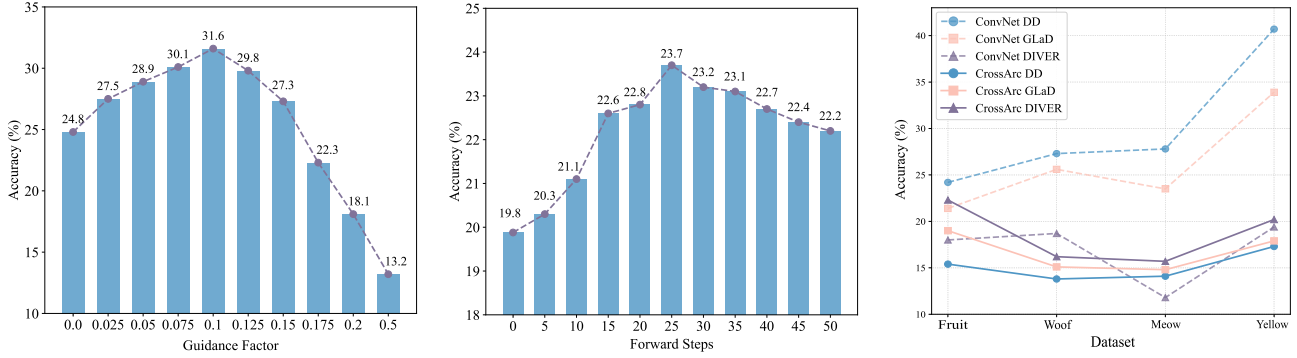


Figure 4. (Left) The effect of guidance factor on performance on ImageYellow (IPC=10) with EDF. (Medium) The effect of applying different forward steps to the inherited latent on performance on ImageFruit (IPC=10) with NCFM (only SI). (Right) Performance of DD (MTT), GLaD, and our DIVER under IPC 1 on the specific ConvNet and across heterogeneous architectures.

Table 7. Sensitivity of our DIVER to different diffusion models. The results are on the ImageNette with MTT.

IPC	DD	Ours		
		SD-V1.5	DiT	SiT
1	17.7 $\pm$ 1.9	19.1 $\pm$ 1.3	20.3 $\pm$ 1.5	20.2 $\pm$ 1.7
10	20.7 $\pm$ 1.8	26.5 $\pm$ 2.1	34.3 $\pm$ 1.4	33.1 $\pm$ 2.3

Table 8. Performance of reconstructing distilled images using only VAE. The results are on the ImageMeow with IPC=10.

Method	Arc.	Distilled	Reconstructed	Ours
DM	ConvNet	23.2 $\pm$ 0.8	21.2 $\pm$ 1.2	19.6 $\pm$ 0.9
	CrossArc	17.6 $\pm$ 1.7	18.9 $\pm$ 1.0	20.0 $\pm$ 1.6
MTT	ConvNet	37.1 $\pm$ 1.3	31.6 $\pm$ 0.7	28.3 $\pm$ 1.1
	CrossArc	16.1 $\pm$ 1.4	21.8 $\pm$ 1.9	26.7 $\pm$ 1.7

Role of VAE. As shown in Tab. 8, the reconstructed images also demonstrate promising generalization capability, exhibiting only a slight performance drop on the prior ConvNet. In contrast, our synthetic images achieve further improvement in cross-architecture generalization, while their distillation performance on ConvNet decreases significantly, reflecting a structured trade-off. Together with the visualization results in Fig. 7 of the appendix, we observe that the architecture-specific patterns consist of **noise coverage** and **semantic degradation**. The VAE primarily filters out “noise”, whereas the diffusion model recovers semantic, and their joint effect enhances generalization performance.

4.4. Analysis

Performance on ConvNet. In Fig. 4 (right), we show the performance on the backbone ConvNet. DIVER shows a measurable decline while maintaining competitive accuracy. We consider this an acceptable trade-off given its superior generalization. This decline primarily stems from two factors: (1) The encoder maps the distilled images into a deep latent space, filtering out architecture-specific patterns (ConvNet) while preserving informative semantics. This is evidenced by our experimental finding in Tab. 8 that directly decoding these latent representations (without diffusion model) still yields considerable generalization gains. (2) The diffusion model further eliminates residual architecture-specific information (degraded semantics) in the latent space through denoising steps, thereby injecting clear semantic information to enhance generalization.

Visualization. As shown in Fig. 5 (left), we compare image generation results from DD and their integration into DIVER on ImageFruit. DIVER distills semantic information from distilled images rich in specific patterns, yielding more realistic and informative images with comprehensive semantic coverage and similar semantic representation.

As shown in Fig. 5 (right), we compare synthetic images generated with (semantic-phase fusion) and without (full-phase fusion) our proposed SF on ImageMeow. SF makes the category information clearer and has better fidelity, while its absence leads to blurring and artifacts in synthetic images, which also accounts for the performance gap described in Tab. 6. This phenomenon primarily stems from two factors: (1) The latent code of the distilled dataset still retains residual architecture-specific patterns, where overfitting exacerbates image distortion and artifacts, inevitably degrading performance. (2) Full-phase guidance obstructs the injection of conditional category information and also runs the risk of overfitting, ultimately compromising visual fidelity.

The t-SNE in Fig. 6 shows that both distilled and reconstructed images are locally clustered and compact, whereas our synthetic images exhibit greater diversity within the original distribution, which is beneficial for generalization. Notably, our synthetic images lie between the broader real-data distribution and the more compact distilled-data distribution, which is consistent with the prior finding (Lee & Chung, 2024) that incorporating a certain amount of real images helps pull the distilled distribution closer to the original one.

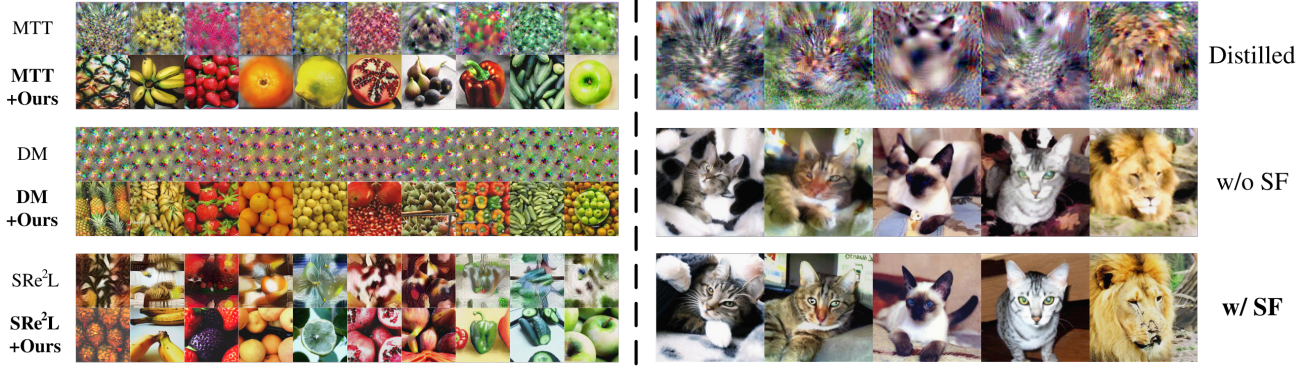


Figure 5. (Left) Comparison of synthetic images from different methods with and without our DIVER on ImageFruit. Our approach recovers expressive semantics and is more realistic. (Right) Comparison of images generated with (semantic-phase fusion) and without (full-phase fusion) our SF on ImageMeow. SF enhances category clarity and fidelity.

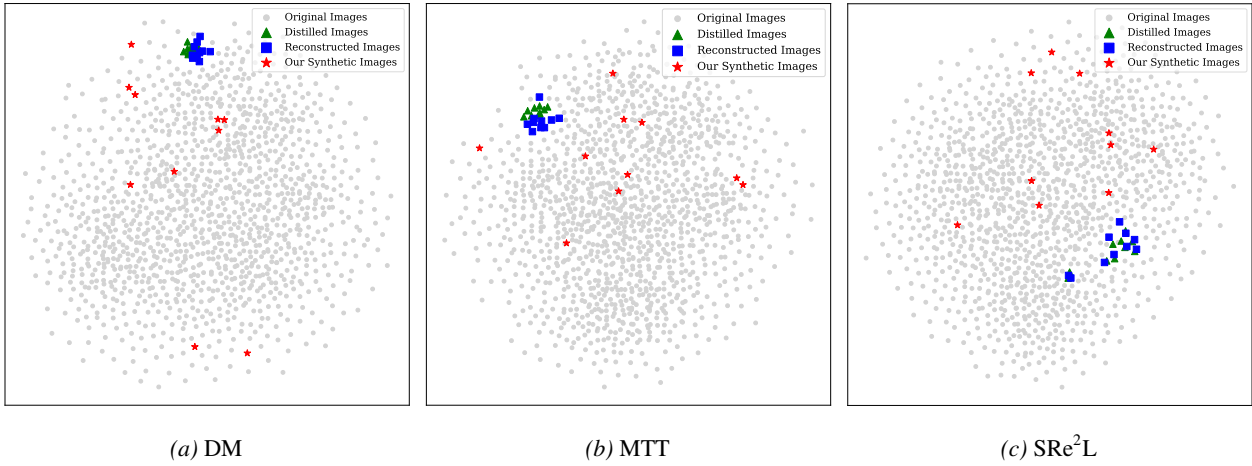


Figure 6. t-SNE visualization of original images, distilled images, reconstructed images (obtained through direct VAE encoding and decoding without the diffusion), and our synthetic images. The results are presented for the first class of ImageNet under 10 IPC. Unlike the distilled and reconstructed samples that tend to form dense local clusters, our synthetic images cover a broader range of the original data distribution, leading to richer variability and improved generalization ability.

Computational and Memory Costs. Our synthetic images achieve superior generalization performance without incurring additional computational costs for fine-tuning. Notably, the architecture-free nature of DIVER guarantees fixed time and GPU memory requirements during synthesis. For our method, SI only needs to encode distilled images into latent codes, incurring negligible computational overhead. SG employs Eqn. 8 to compute gradients directly in the latent space for sampling guidance, with this additional operation introducing only minimal computational burden. To further minimize the computational cost, we set $\nabla_{\hat{z}_t} \mathcal{G}_t(\hat{z}_t) = (\hat{z}_t - z_0)\sigma_t$, eliminating gradient calculations entirely during sampling. Combined with SF, our approach achieves sampling time comparable to raw DiT (2.41s per image). Our method takes only 2.48s per image processing on ImageNet (256×256) with 4.02 GB memory usage on a single RTX-4090 GPU, demonstrating the computational and memory efficiency of our framework.

5. Conclusion

In this paper, we propose a novel task of “Diving into Distilled Data” for the first time, where we enhance remarkable cross-architecture generalization by recovering the expressive semantics suppressed by specific patterns in the distilled dataset. Our method can be used as a plug-in to directly optimize the distilled dataset generated by classical dataset distillation in a raw data-free and training-free manner. This efficient dual-stage paradigm achieves high model performance with minimal storage and computational overhead.

Limitation and Future Work. Currently, our method is heavily dependent on the quality of the distilled images and has only been extended to a limited range of diffusion-based approaches. In future work, we plan to directly integrate the informative patterns from conventional distilled images into other diffusion-based frameworks to further improve the scalability and generalization.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Appendix

Algorithm 1: DIVER: Diving Deeper into Distilled Data via Expressive Semantic Recovery

Input: Pre-trained diffusion model ϵ_θ , VAE encoder $\mathcal{E}$ and decoder $\mathcal{F}$, distilled dataset $\mathcal{D}^*$
(or original dataset $\mathcal{O}$ and classical dataset distillation algorithm Alg)

Output: Synthetic dataset $\mathcal{S}$

Params: Class c , classifier-free guidance (cfg) factor w , semantic guidance factor γ , guided range t_l, t_h , diffusion steps t_f, t_r , scales $\{\alpha_t\}_{t=1}^{t_f}$

```

1 Obtain the distilled dataset  $\mathcal{D} = \mathcal{D}^*$  if  $\mathcal{D}^*$  exists else  $\mathcal{D} = Alg(\mathcal{O})$ ;
2 for each image  $x_0 \in \mathcal{D}$  do
3   Perform Semantic Inheritance:  $z_0 = \mathcal{E}(x_0)$ ;
4   Sample Gaussian noise:  $\epsilon \sim \mathcal{N}(0, I)$ ;
5   Perform the forward process as Equation 7:  $\hat{z}_{t_r} = z_{t_f} = \sqrt{\alpha_{t_f}}z_0 + \sqrt{1 - \alpha_{t_f}}\epsilon$ ;
6   for  $t = t_r$  down to 1 do
7     Obtain the predicted cfg noise:  $\epsilon_\theta = \epsilon_\theta(\hat{z}_t, t, \emptyset) + w \cdot (\epsilon_\theta(\hat{z}_t, t, c) - \epsilon_\theta(\hat{z}_t, t, \emptyset))$ ;
8     if  $t_l \leq t \leq t_h$  then (Semantic Fusion)
9       Compute the guidance as Equation 8:  $\nabla_{\hat{z}_t} \mathcal{G}_t(\hat{z}_t) = (\hat{z}_t - z_0)\sigma_t$ ;
10      Execute Semantic Guidance:  $\hat{z}_{t-1} = s(\hat{z}_t, t, \epsilon_\theta) - \gamma \cdot \nabla_{\hat{z}_t} \mathcal{G}_t(\hat{z}_t)$ ;
11    else
12      Execute vanilla sampling:  $\hat{z}_{t-1} = s(\hat{z}_t, t, \epsilon_\theta)$ ;
13 return Decoded synthetic dataset:  $\mathcal{S} = \{\mathcal{F}(\hat{z}_0)\}$ ;
    
```

A.1. Related Work

A.1.1. DATASET DISTILLATION

Dataset Distillation (DD) condenses a large original dataset into a compact distilled dataset, preserving essential information to maintain comparable test performance when training models. These informative images are also valuable for various applications, including continual learning (Yang et al., 2023; Gu et al., 2023), federated learning (Yan et al., 2025; Huang et al., 2024), and neural architecture search (Ding et al., 2024; Such et al., 2020). Moreover, DD techniques are being increasingly applied to a wider range of domains, including graph neural networks (Liu et al., 2024b; Fang et al., 2026a), long-tailed learning (Zhao et al., 2025b; Zhang et al., 2023), and domain adaptation (Montesuma et al., 2024; Fang et al., 2026b; 2025). The classical DD framework utilizes bi-level optimization for dataset generation. This involves an inner loop that updates the network to assess classification performance, while the outer loop synthesizes images through specific matching strategies. Various DD approaches have emerged from this framework, including gradient matching (Loo et al., 2023; Wang et al., 2023), distribution matching (Wang et al., 2022a; Deng et al., 2024; Zhang et al., 2024), and trajectory matching (Du et al., 2023; Zhong et al., 2025; Guo et al., 2023). Inspired by Data-Free Knowledge Distillation (DFKD) (Liu et al., 2024a; Chen et al., 2019), recent studies propose decoupled dual-time optimization approaches (Yin et al., 2023; Shao et al., 2024) that separate the distillation process into synthesis and training phases, enabling more efficient handling of large datasets. The methods alleviate some constraints on visual semantics in distilled images.

Although the images synthesized by traditional methods are insightful and this expressiveness may enhance distillation performance, all perform the optimization process directly in the pixel space, which tends to excessively learn specific patterns that overfit on a prior architecture (Cazenavette et al., 2023). The resulting distilled images exhibit *abstract*, *noisy*, and *unrealistic* characteristics. The visual semantics with realism that help generalize across architectures are suppressed (Sun et al., 2024). Therefore, generation-based methods emerged.

A.1.2. DATASET SYNTHESIS WITH GENERATIVE MODEL

Generative prior methods (Cazenavette et al., 2023; Zhong et al., 2024) use GANs (Karras et al., 2019; 2020) to synthesize images, transitioning the optimization space from pixels to latent codes. But they still rely on the traditional DD paradigm, and their effectiveness remains constrained by expensive inner loop matching mechanisms and inadequate semi-realistic representation. Consequently, several diffusion-based approaches (Su et al., 2024; Gu et al., 2024; Chan-Santiago et al.,

Table 9. Distillation performance on the ConvNet architecture.

Distill. Alg.	IPC	Distill. Mode	Fruit	Woof	Meow	Squawk	Nette	Yellow
DM	1	DD	19.5±1.0	20.2±0.6	18.3±1.7	27.3±0.8	28.9±1.5	32.9±1.6
		GLaD	18.9±1.7	18.2±1.9	20.5±2.1	23.2±1.6	30.3±1.1	30.3±1.0
		DIVER	13.2±1.0	13.6±0.6	10.5±1.3	18.2±0.7	21.6±1.2	17.8±0.7
	10	DD	26.2±0.5	24.0±0.6	23.2±0.8	32.7±1.2	40.0±1.1	41.5±0.6
		DIVER	18.4±0.7	23.1±0.8	19.6±0.9	31.2±1.7	35.0±1.7	32.0±0.4
MTT	1	DD	24.2±0.7	27.3±1.0	27.8±1.1	12.0±1.6	17.7±1.9	40.7±2.9
		GLaD	21.4±1.2	25.6±1.4	23.5±1.8	13.6±1.2	22.5±1.4	33.9±1.4
		DIVER	18.0±0.4	18.7±1.5	11.8±0.9	11.6±1.0	20.3±1.5	19.4±0.6
	10	DD	35.3±1.4	32.9±0.8	37.1±1.3	49.9±1.8	20.7±1.8	53.8±0.8
		DIVER	25.7±0.8	25.5±0.8	28.3±1.1	39.1±1.0	34.3±1.4	37.5±1.3

2025; Chen et al., 2025; Zhao et al., 2025a) leveraging the powerful synthesis capabilities of diffusion models have begun to emerge and demonstrated promising performance. These methods mark a radical break from the traditional DD paradigm, completely abandoning its established norms in order to refocus squarely on the core principles of coreset construction. Some novel strategies (such as prototype learning, influence function, etc.) are used to synthesize representative samples.

Inspired by both single-stage classical DD and generative-based approaches, we implement our work by integrating the expressive knowledge distilled from traditional DD with the powerful realistic image generation capability of diffusion models. This hybrid framework aims to synthesize a visually and semantically enriched dataset, thereby enhancing cross-architecture generalization.

A.1.3. IMAGE RESTORATION WITH GENERATIVE MODEL

Image restoration is a fundamental discipline within image processing, concerned with the inverse problem of reconstructing a high-quality image from its degraded observation. The field encompasses several key sub-domains, each targeting specific types of degradation, such as image deblurring (Chen et al., 2024), super-resolution (Yue et al., 2025), and adverse weather removal (Peng et al., 2025), including tasks like de-raining, de-snowing, and de-hazing to eliminate weather-related artifacts.

Recently, some image restoration techniques (Yue et al., 2025; Wu et al., 2024; Wang et al., 2024) based on diffusion models have begun to attract researchers’ attention. Although similar to DIVER, both aim to synthesize realistic and semantically clear images using prior knowledge from generative models, they are fundamentally different. (1) Restoration tasks typically require ground-truth data (clean images) for training. In contrast, our method does not rely on any other images and does not require training. (2) Restoration seeks to recover a clean natural image from a degraded observation (e.g., blurring, noise, low resolution). Our distilled images are not degraded natural images, but optimization artifacts containing architecture-specific patterns, so the objective is semantic refinement rather than reconstruction.

A.2. The Theoretical Explanation of The Gain Mechanism

A key limitation of classical distilled datasets is their insufficient generalization ability, mainly due to: (1) **overfitting to architecture-specific high-frequency “noise”**, and (2) **image artifacts and abstraction caused by pixel-level optimization**. Our VAE (SI) is used to solve (1), and SG and SF are used to solve (2), as explained below:

1. Why VAE (SI) suppresses high-frequency “noise”

VAE performs lossy compression: high-frequency “noise” that is hard to explain and inconsistent with the prior is treated as “non-essential” and gets discarded or averaged out. The standard VAE optimizes the ELBO:

$$\max \mathbb{E}_{q(z|x)}[\log p(x|z)] - \beta \text{KL}(q(z|x)||p(z))$$

The first term encourages accurate reconstruction of x from z . The second term pushes the posterior $q(z|x)$ to stay close to the prior $p(z)$ (often a standard Gaussian). For high-frequency “noise” of distilled images, encoding them precisely often requires $q(z|x)$ to become sharp or deviate from Gaussian along some dimensions, which **increases the KL**. But their contribution to “generalizable semantics” is small (and may even be harmful for downstream learning), so under the optimal trade-off, the model **tends not to encode them**. As a result of Fig. 7 (right), the VAE drops them as “high cost and low

Table 10. Effect of DiT models with different resolutions on cross-architecture generalization on ImageNet Subsets.

Resolution	Fruit	Woof	Meow	Squawk	Nette	Yellow
256×256	22.3	16.2	15.7	17.2	20.3	20.2
512×512	22.4	17.1	17.8	17.4	23.3	20.5

Table 11. The synthesis time and GPU memory cost of different devices on ImageNet.

Device	Resolution	GPU (GB) ↓	Times (s) ↓	
			DiT	Ours
4090	256×256	4.02	2.41	2.48
	512×512	5.44	12.84	13.08
A800	256×256	4.02	1.92	1.97
	512×512	5.44	6.74	6.91

benefit” information, and the decoded image naturally looks cleaner. As shown in Tab. 8, images directly reconstructed by the VAE lead to lower ConvNet performance but better generalization, which further supports our conclusion.

2. Diffusion enhances fidelity while preserving information

Images reconstructed by the VAE remain abstract and unrealistic. They often deviate from the true manifold (see Fig. 6), thus impairing generalization. Therefore, the role of diffusion is to bring the reconstructed images back to the real data manifold while preserving the valuable information encoded by the VAE in (1). For **real data manifold**, this is determined by the characteristics of the diffusion-projected noise to the real data distribution. For **preserving the valuable information**, our SG encourages the denoised latent to move toward the VAE-encoded latent in (1), thereby preserving the necessary information $\mathcal{G}_t = (\hat{z}_t - z_0)^2$. SF further prevents overfitting to such information beyond the real data distribution, thereby preserving image fidelity (see Fig. 5 right). In this way, our entire pipeline removes the architecture-specific noise and abstract representations in classical distilled images while preserving valuable semantic information.

A.3. More Experiments

Quantitative Results on ConvNet. As shown in Tab. 9, we compare the distillation performance of DM and MTT on ConvNet under DD and DIVER. In most cases, the performance of our method does drop significantly. We have analyzed the reasons in the text. This performance decline primarily stems from two factors: (1) The encoder maps the distilled images into a deep latent space, filtering out specific patterns (ConvNet) while preserving high-level semantics. This is evidenced by our experimental finding that directly decoding these latent representations (without employing the denoising model) still yields considerable generalization gains. (2) The diffusion model further eliminates residual low-level information in the latent space through denoising steps, thereby injecting additional class-specific information to enhance generalization. Notably, DIVER with MTT still achieves performance gains on the ImageNette dataset. This may be because the distilled images contain excessive invalid “noise”, which not only suppresses semantic expression but also hinders distillation performance. In contrast, diffusion filters out this portion of ineffective low-level information, thereby improving distillation performance. This observation also inspires our future work, extracting meaningful low-level features to further improve distillation performance on ConvNet.

Effect of Different Resolutions. We compare the effect of DiT models with different resolutions on cross-architecture generalization in Tab. 10. As the resolution increases, the performance improves slightly, which may be because the synthesized images better inherit the semantic details of the distilled images. However, as shown in Tab. 11, this is accompanied by a significant increase in synthesis time. And better equipment can also improve the synthesis efficiency.

Table 12. Integrate MGD³ into DIVER in different ways. The results are obtained by averaging three experiments each on ResNet-18, ResNetAP-10 and ConvNet-6 on ImageNette with IPC 10.

Method	Original	Synthesis-based	Prototype-based
MGD ³	60.3±0.3	51.7±0.5	62.8 ±0.4

Table 13. Comparison of optimized DD (NCFM) with diverse initializations versus our DIVER on ImageSquawk under IPC=1.

Init.	Mode	ResNet18	ShuffleNet-V2	EfficientNet-B0	ViT-b/16
Random	DD	21.2 $\pm$ 1.4	18.0 $\pm$ 3.3	14.1 $\pm$ 2.1	17.2 $\pm$ 0.6
	Ours	23.1$\pm$2.5	19.6$\pm$1.8	18.7$\pm$1.9	23.6$\pm$0.4
Noise	DD	14.4 $\pm$ 1.9	15.6 $\pm$ 2.2	11.0 $\pm$ 1.8	19.4 $\pm$ 1.0
	Ours	20.8$\pm$2.6	21.2$\pm$3.4	15.5$\pm$0.9	21.7$\pm$1.0
Mix	DD	17.6 $\pm$ 1.5	15.3 $\pm$ 2.0	13.6 $\pm$ 0.2	18.9 $\pm$ 1.8
	Ours	22.6$\pm$0.8	18.5$\pm$2.0	14.2$\pm$2.2	19.8$\pm$1.1

Table 14. Effect under high IPC settings.

IPC		10	50	100
ConvNet	DM	23.2 $\pm$ 0.8	22.7 $\pm$ 1.1	20.9 $\pm$ 2.1
	Ours	19.6$\pm$0.9	24.8$\pm$0.3	28.5$\pm$1.5
CrossArc	DM	17.6 $\pm$ 1.7	16.4 $\pm$ 1.5	16.8 $\pm$ 1.3
	Ours	20.0$\pm$1.6	26.5$\pm$2.0	32.6$\pm$1.8

Synthetic Images for Generative Models. As shown in Tab. 12, we apply the synthetic dataset generated by MGD³ directly to our framework, which reduces performance by approximately 10% (51.7% vs. 60.3%). Because the synthetic images contain relatively clear semantic structures. When they are used in our framework, the performance degradation is mainly attributed to (1) the encoding of VAE from high-dimensional to low-dimensional space itself loses information, and (2) uncertainty introduced by diffusion. Prototype-based methods retain valuable information from the original dataset or prototype, thereby achieving performance improvements.

Initialization of Distilled Images. To assess the robustness of DIVER, we obtain distilled datasets optimized under different initial image settings, including: (1) random selection from original images, (2) Gaussian noise, and (3) mixed images composited from four source images, and all integrated into our framework. As shown in Tab. 13, our method consistently improves the generalization of distilled datasets optimized under various conditions, demonstrating its superiority and stability.

Effect under high IPC settings. In practical applications, we place greater emphasis on the generalizability of information contained within distilled datasets to facilitate their use in training or fine-tuning other architectures, rather than being confined to a specific architecture like ConvNet. As illustrated in Tab. 14, datasets distilled through classical DD exhibit unstable performance under 50 IPC and 100 IPC. Traditional DD struggles significantly, with no improvement in generalization as IPC increases, even showing degradation under ConvNet evaluation. In contrast, our method demonstrates consistent gains, validating the significance of capturing universal information from original images rather than architecture-specific features to enhance generalization.

Visualization. As shown in Fig. 8, different images also exhibit distinct optimal noise-addition steps. For each image, as the number of forward steps increases, the feature variations (e.g., texture details) gradually intensify. However, this change eventually diminishes and stabilizes, as the initial latent variables asymptotically converge to the same Gaussian distribution. Our visual comparison of MTT and DIVER under IPC=10 is presented in Fig. 9 ~ Fig. 14. Our method releases the expressive semantics that are suppressed in raw DD, and the valuable label semantic information covers almost the entire image.

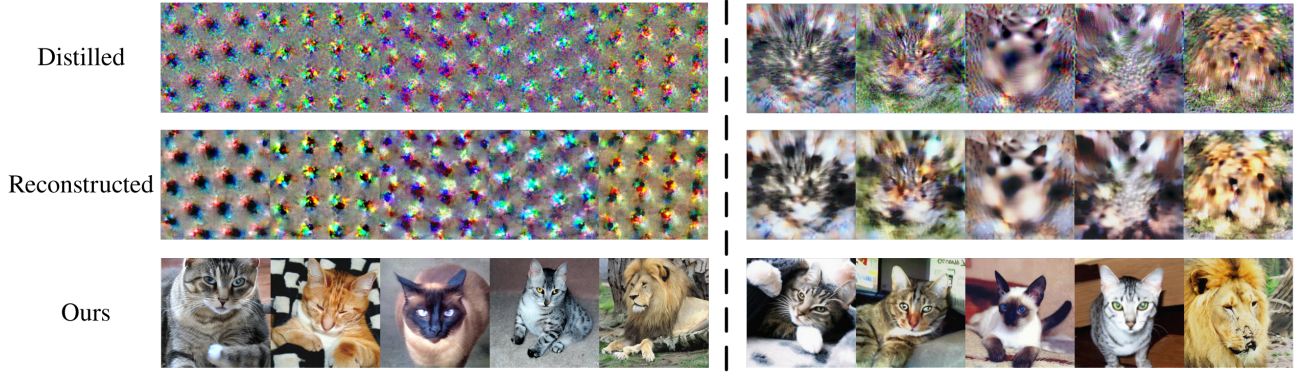


Figure 7. Comparison of distilled images, reconstructed images (obtained through direct VAE encoding and decoding without DiT), and our synthetic images with DM(left) and MTT(right).

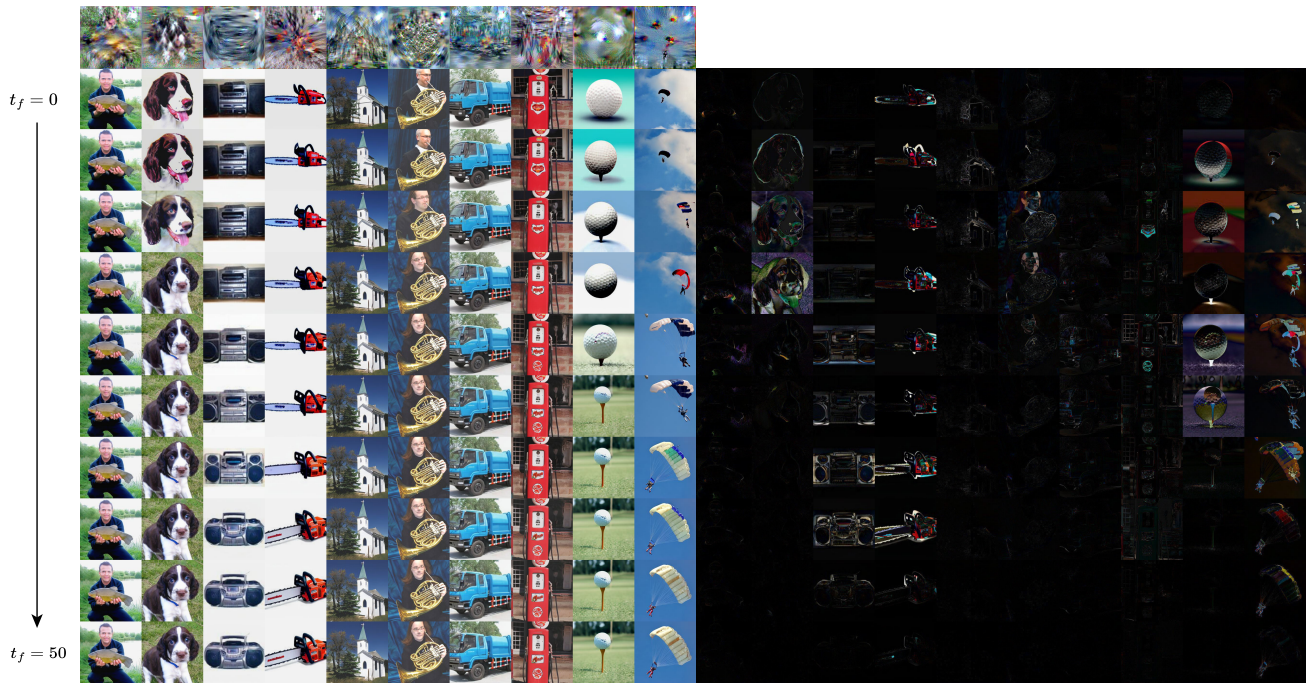


Figure 8. The pixel difference between the images synthesized by adjacent forward steps.



Figure 9. Visualization comparison between raw DD and DIVER on ImageFruit.



Figure 10. Visualization comparison between raw DD and DIVER on ImageWoof.

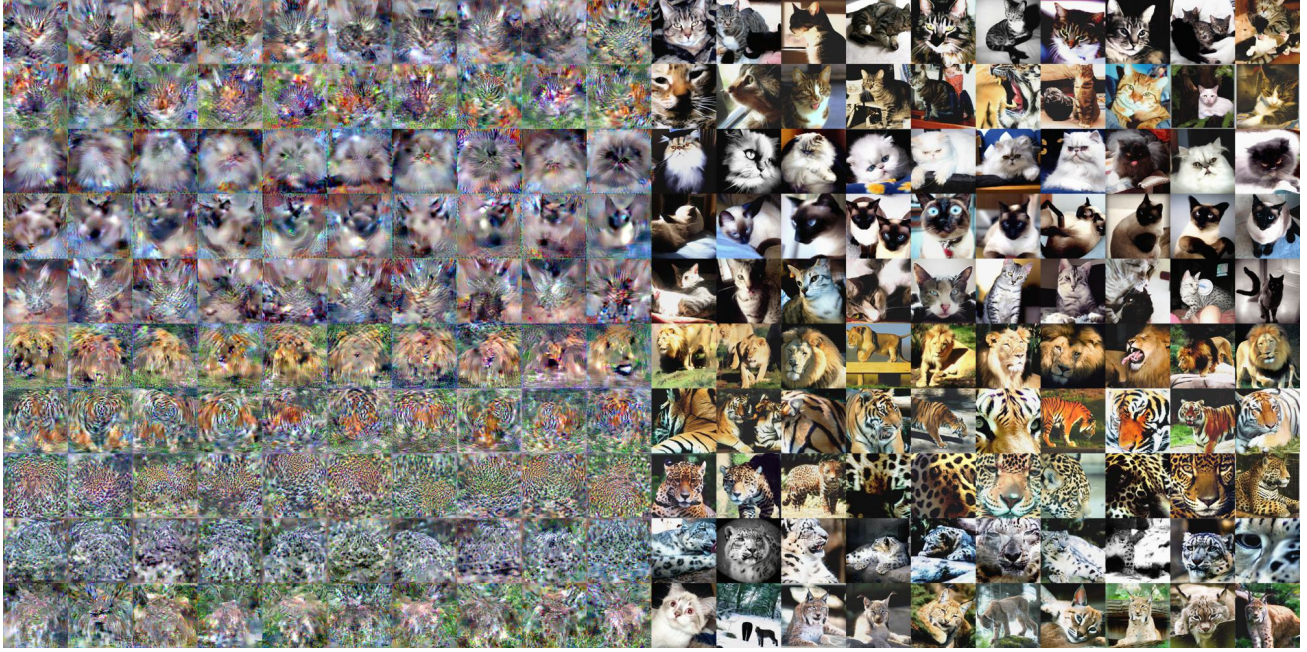


Figure 11. Visualization comparison between raw DD and DIVER on ImageMeow.

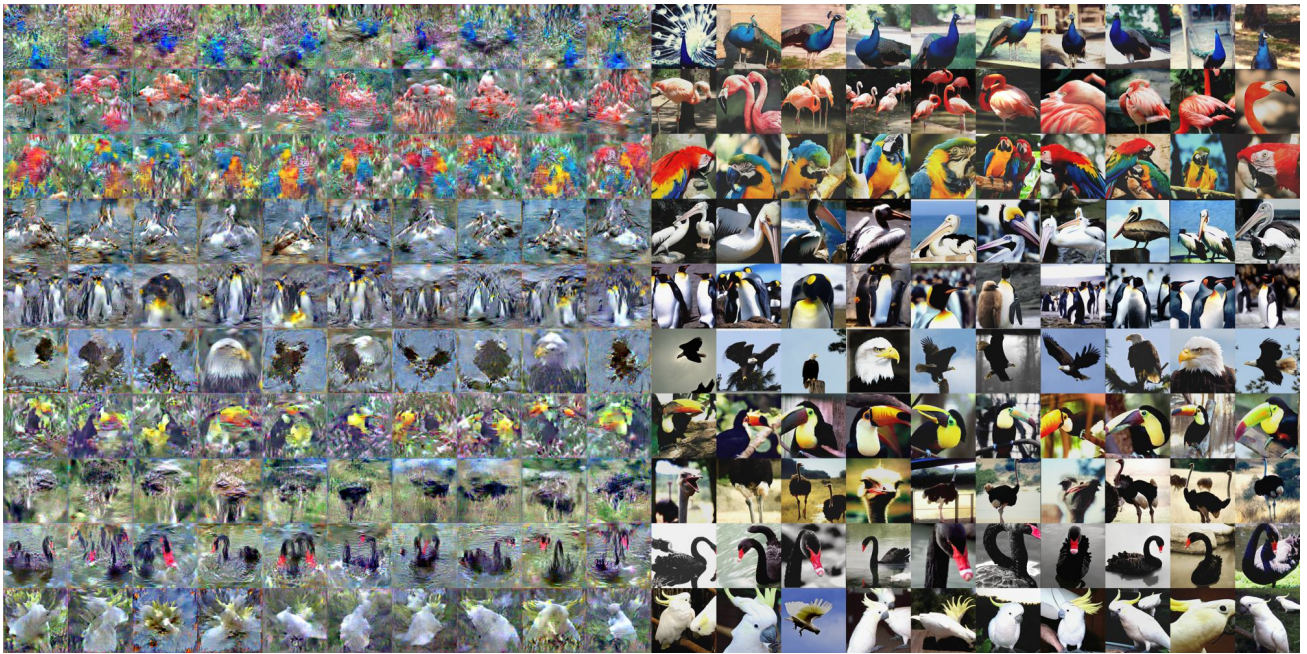


Figure 12. Visualization comparison between raw DD and DIVER on ImageSquawk.



Figure 13. Visualization comparison between raw DD and DIVER on ImageNette.

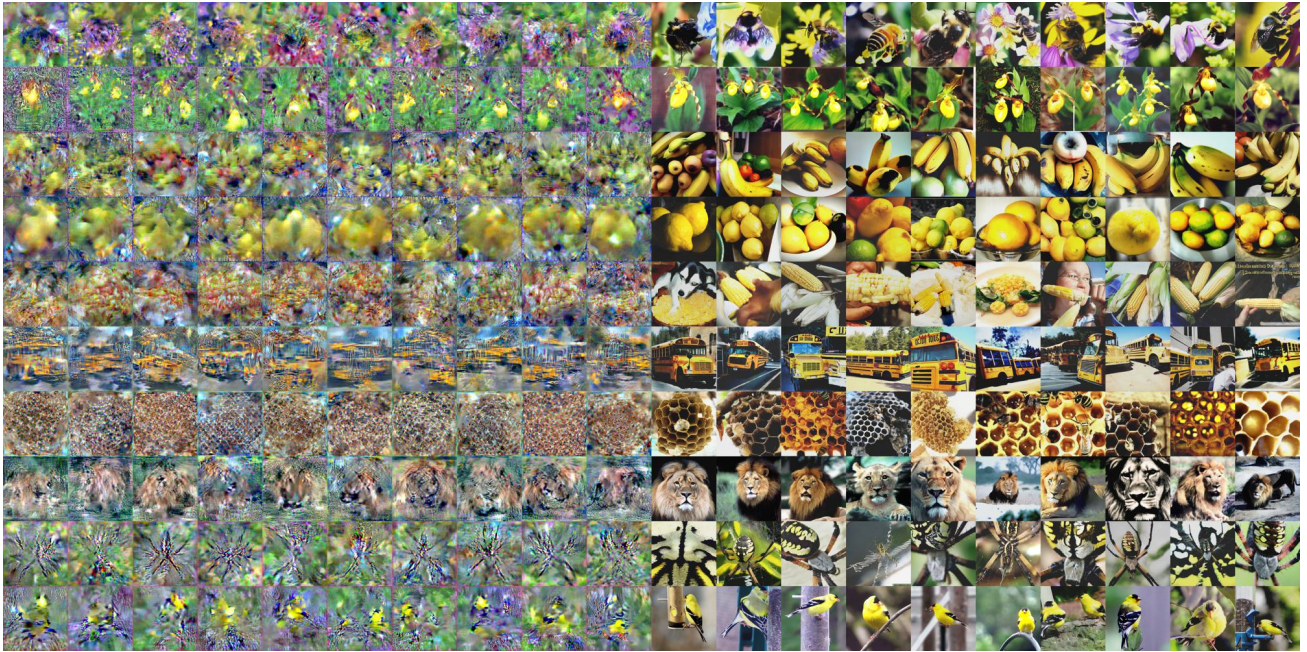


Figure 14. Visualization comparison between raw DD and DIVER on ImageYellow.